

Evaluation of Sharknado!

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Table of contents

1 Executive Summary

1.1 Purpose of Report

This study dives into the possibility of a Sharknado event. A Sharknado can be explained as a “shark-infested tornado”. This is a tornado that generates enough wind speed to lift a great white shark from the water and into its vortex. The purpose of this study is to assess whether a Sharknado event could be considered a risk for property damage insurance payouts.

1.2 Methods

To start, a comparison of the weight of a great white shark (around 4500 lbs) and vehicles (around 4000 - 4600 lbs for SUVs and Trucks) is used because cars are known to be lifted and destroyed by tornadoes. Using the EF (Enhanced Fujita) Scale, the strength of tornadoes able to lift and toss cars (or a great white shark) is found to be between EF3 - EF5 tornadoes. These are wind speeds of 136 mph to 200+ mph. This study will only be looking at tornadoes of EF3 - EF5 for this reason. This study also looks at four locations of tornado data from 2015 to 2026. These locations are Florida, South Carolina, Virginia, Oklahoma, California, and Los Angeles County, CA. These places were chosen because of their abundant tropical storms and coastal area (Florida and South Carolina), historic tornado amounts and strength (Oklahoma), and the setting of the movie Sharknado (Los Angeles, CA).

1.3 Findings

Tornadoes of strength EF3 - EF5 are fairly uncommon for coastal states like Florida and South Carolina. Florida had only five tornadoes in the last 10 years that were of strength EF3, and zero EF4 and EF5 tornadoes. All but one of these tornadoes started and ended on land. The one tornado started on a beach in Florida, where the water is not deep enough for sharks. South Carolina had nine EF3 and one EF4 tornado over the last 10 years. All of these tornadoes started and stayed on land, making a Sharknado impossible. Virginia had five EF3 tornadoes that all happened on land. Oklahoma has had plenty of tornadoes strong enough to lift and carry a shark, but there are no sharks in Oklahoma besides those in the Oklahoma Aquarium in Jenks, OK. California has not had a tornado above an EF2 over the last 10 years.

1.4 Conclusions

The probability of a Sharknado event is extremely low, if not a zero percent chance. There was one tornado in the three coastal states that might have had the ability to lift a shark. This South Carolina tornado happened to start and end on land, making a Sharknado impossible. The Oklahoma Aquarium houses sharks in a state that is known for its number and strength of tornadoes. The Oklahoma Aquarium is in Tulsa County, OK, where only EF2 or lower tornadoes have touched down in the last 10 years, rendering this very improbable. From the findings of this study, the probability of a Sharknado happening is almost, if not, a zero percent chance.

1.5 Recommendations

From this study, Sharknados are certainly extremely unlikely, if not a zero percent chance. Even though strong storms and tornadoes without sharks are still a cause for concern. Property damage and its resulting insurance payouts will still be present with storms and tornadoes that have high wind speeds and extreme precipitation events. These are real and worrisome outcomes that happen all over the country that are more probable than a Sharknado and should be taken more into account.

2 Introduction

This reports evaluates the potential insurance risk associated with a Sharknado scenario for property damage payouts . A Sharknado is explained by a tornado that generates enough wind speed to be able to lift a heavy animal, such as a great white shark, out of the water and into its vortex. A Sharknado can also be called a “Shark-infested tornado”.

This study starts by comparing the average great white shark to objects of similar weight, specifically cars or pickup trucks. Tornadoes are known for destroying homes and tossing around vehicles, thus finding the types of vehicles that are similar enough in weight to a great white shark makes finding the right magnitude of tornadoes required to lift a great white shark easier.

Next, finding the magnitude of a tornado capable of picking up a great white shark, or a car, is found using the EF scale (Enhanced Fujita Scale) by looking at EF ratings (US Department of Commerce (n.d.)). The Enhanced Fujita Scale uses ratings of EF0 to EF5. Tornadoes of interest for this study are tornadoes of rating EF3 to EF5. These tornadoes have strong enough wind speeds to potentially pick up and throw cars or, in this case, a great white shark.

Finally, looking at places that are near the coast to be able to pick up a great white shark is essential to actually generating a Sharknado. There are 4 locations observed for their tornado data that fit this agenda in this study: Florida, South Carolina, Virginia, California, and Los Angeles (this is the setting of the Sharknado movie). For another comparison of tornado strength and amount, Oklahoma is also looked at for its tornado EF rating, even though it is landlocked and does not have any native shark population. These locations have tornado data taken from 2015 - 2026 using NOAA tornado data.

Combining the magnitude of tornadoes with the areas where these tornadoes touch down gives an idea of the possibility of a potential Sharknado event occurring.

3 Data Documentation

All data was collected from NOAA (National Oceanic and Atmospheric Administration) Storm Events Database. The NOAA's storm events database contains records on various types of severe weather collected by NOAA's National Weather Service.

Tornado data from NOAA Storm Events Database was collected for five different states and one county across the United States ("Storm Events Database | National Centers for Environmental Information" (n.d.)). These locations were chosen because of their amount of EF3 or higher tornadoes, location close to the ocean, and setting of the Sharknado movie. These locations are:

- Florida: high frequency of tornadoes EF3 or higher for a coastal region
- South Carolina: high frequency of tornadoes EF3 or higher for a coastal region
- Virginia: high frequency of tornadoes EF3 or higher for a coastal region
- Oklahoma: high amount of powerful tornadoes to compare the frequency of high magnitude tornadoes
- California and Los Angeles, CA: State and city setting of the movie "Sharknado" ("Sharknado (TV Movie 2013) - Filming & Production - IMDb" (2013))

All data was analyzed using the programming software R and these following R packages:

- `tidyverse`: data cleaning, data manipulation, and table formulation
- `gt`: to make better looking tables
- `maps`: make maps of United States for tornado data

4 Methods

A Sharknado event involves a tornado being able to lift a great white shark out of the ocean and carry it onto land. To see if this is possible, we need to look at the physics: the weight of a great white shark, what objects are known to be picked up by tornadoes, and the magnitude of tornadoes required to pick up a shark. First, finding out how a great white shark could be picked up by a tornado is necessary. The weight of a great white shark will be compared to other objects that can and have been picked up by tornadoes, such as cars. Next, figuring out what strength (in EF scale from EF3 - EF5) of a tornado would be needed to be able to lift a great white shark and a car. Lastly, identifying tornadoes from 2015-2026 that are in that EF range from 6 different places in the United States. These places are Florida, South Carolina, Virginia, Oklahoma, California, and Los Angeles. These places were chosen for specific reasons. Florida has many tropical storms and is a coastal state (Florida, (n.d.)). South Carolina is also another place that can be hit by tropical storms and is another coastal state (South Carolina, (n.d.)). Virginia is a coastal state that has had a good amount of tornadoes of our criteria (Virginia, (n.d.b)). Oklahoma is not a coastal state, but it is the tornado capital of the United States, so we will just look at the number of tornadoes that meet the criteria to get an idea of how rare strong tornadoes are (Oklahoma, (n.d.)). California ((n.d.a)) was chosen because it is the state setting of Sharknado, while Los Angeles ((n.d.)) was picked because it is the city setting of the Sharknado movie ("Sharknado (TV Movie 2013) - Filming & Production - IMDb" (2013)).

Sharks and cars weigh relatively the same in pounds (lbs). The average great white shark weighs around 4,500 lbs (Fisheries (2025)). A car SUV weighs around 3,986 lbs while a truck SUV weighs around 4,578 lbs (AutoInsurance (n.d.)). This means that your average great white shark weighs more than a big car and as much as a truck or SUV. We will use a car as the base to find what EF scale level will be used to lift a great white shark out of the water.

The EF scale is made up of 6 different ratings of tornado based on 3-second wind speed in mph (US Department of Commerce (n.d.)). An EF0 tornado has the lowest wind speed, ranging between 65-85 mph, EF1 has wind speeds between 86-110 mph, EF2 has wind speeds between 111-135 mph, EF3 has wind speeds of 136-165 mph, EF4 has speeds between 166-200 mph, and EF5 has wind speeds that are over 200 mph. An EF3 tornado is likely to toss cars and trucks, while an EF4 is very likely to toss cars many feet, and an EF5 tornado is almost certain to lift and destroy cars. EF3 tornadoes are unlikely to fully pick up a shark but it is still important to analyze them because a great white shark could be moved by an EF3 tornado. Because of the comparison of a great white shark to the size and weight of a car SUV, and a truck SUV, that gives a good comparison to be able to put a shark in the place of such cars in the EF

scales' ability to lift cars (Quach (2025)). Tornadoes from EF3 to EF5 are chosen because this is the strength of tornadoes that are able to lift a car (Salaita (n.d.)).

Using these metrics, finding storms that have had the potential to produce a Sharknado allows the ability to evaluate the potential insurance risk with such events.

5 Results

The results of this study examines tornado activity over the last 10 years from specific states across the United States. Tornado strength of EF3 - EF5 is chosen due to their capabilities to lift and toss objects that are similar to sharks weight wise, these objects being cars or trucks. The findings are shown by state and area (FL, SC, VA, CA, Los Angeles) to be able to see where and which tornadoes had the strength and location to start a Sharknado event. California and Los Angeles, CA were chosen due to being the location of the film “Sharknado”.

Below are these tornadoes being mapped on the entire United States to get an idea of where these states and their tornadoes are and why they were chosen for this study.

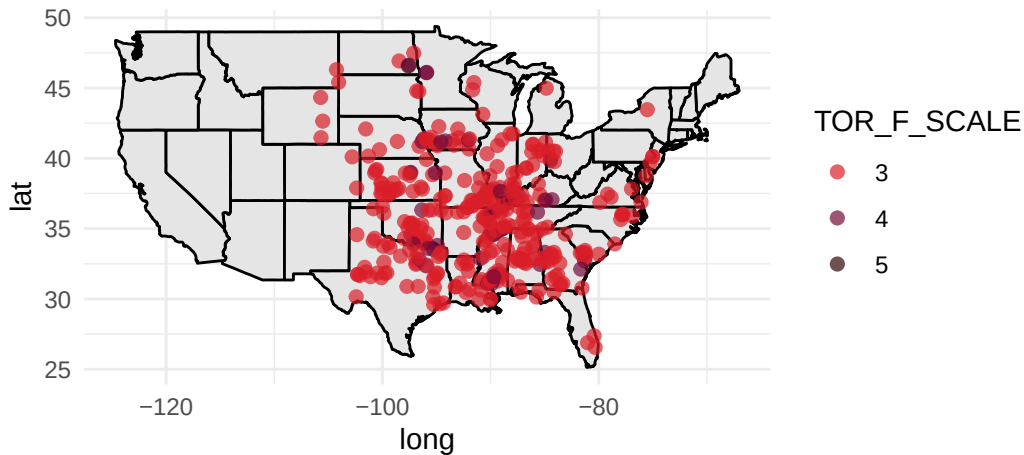


Figure 5.1: EF3 - EF5 Tornado Data Accross USA

Most tornadoes happen on the eastern side of the country as shown in Figure ???. This shows the regions to check to see if Sharknadoes are possible. The east coast is the best place to start the search. The states that have enough tornadoes, and the states that will be examined, are Florida, South Carolina, and Virginia.